24/4/26

Exp (13) Implementation of Ping Program

Aim :-

To write a program to ping a host name as IP address using the net address class and write appropriate result

Procedure :-

1. Open a IDE Java terminal and create a file named program Java
2. Import Inet address and Scanner class
3. Get the hostname as IP address from the user
4. Calculate and display the response time in millisecond.
5. Display reply form

Result :-

The Java program was successfully created the ping given hostname / IP host is reach and shows the response time.

Code :-

```
import socket
import struct
import time
import os

def checksum(data):
    sum = 0
    for i in range(0, len(data)-1, 2):
    sum += (data[i] << 8 + data[i+1])

if len(data) % 2
    sum += (data[i] << 8 + data[i+1])
    sum = (sum >> 16) + (sum & 0xffff)
    sum += sum >> 16
    answer = ~sum >> 16

def Ping(host):-

try:
    dest = socket.gethostbyname(host)
    soc = socket.socket(socket.AF_INET)
    socket.socket(socket.IPPROTO_IMP)

socket.settimeout(2)
Packet_id = os.getpid() & 0xffff

sequence = 1
header = struct.Pack("/BBHHH", 8, 0, 0, Packet_id)
data = struct.Pack("id", time.sequence)
                    time())
```

OUTPUT:

Enter host (eg. google.com): google.com
Pinging 142.250.206.78
reply from 142.250.206.78: time = 80ms
reply from 142.250.206.78: time = 188ms
reply from 142.250.206.78: time = 1019ms
reply from 142.250.206.78: time = 73ms

```
checksum = checksum (header + data)
header = struct.pack ("!BHHH", 8, 0, checksum, Pa
identifier)
Packet = header + data
start = time.time()
while True:
    try:
        ready = sock.recvfrom (1024)
        end = time.time()
        break
    Print ("request timed out")
    continue
delay = (end - start) * 1000
Print ("reply from", address[0])
Print ("time = {:0.2f} ms".format delay))
sock.close()
except Permission error:
    Print ("Please run the Program as administ
except Exception as:
    Print ("error", e)
host = input ("Enter destination:")
Ping (host)
```